An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

Yuya Uezato　Yoshiki Nakamura

CyberAgent, Inc.　Chiba University

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論文 v8 との差分（用語）：

- fold → flip
- notice → (newly) distinguish：同じ色の辺で真偽が食い違う。notice では「気づいてどうなるか」（そのゲートで初めて区別が生まれる）が伝わらない
- reference structure → $\mathcal{M}_0$ ／ read・witness edge → x-y・x-z edge
- color-determined（v8）→ color-invariant。bisimulation-invariant などと同じ「同色の点対で値が同じ」を意図

論文 v8 との差分（内容）：

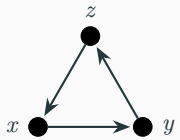
- Key Lemma「d 個を見分ける項は合成 $\geq d/2$」はスライド独自のまとめ方（v8：各 fold はどこかの合成で notice される ＋ 合成 1 つで高々 2）。論文でもこの形で先に述べることを提案
- distinguish を定義として明示（v8 は prediction と notice で定義）
- 省略：$(\hbar, \hbar)$、端点 witness、ループ（p.11 で "see the paper"）。Cor 1.3（CoR 等式）、Cor 1.8・1.9（固定サイズ・乱択・ランダム入力）、項サイズ $(2 + o(1))k2^k$
- 名前のみ：Thm 1.4（FO^3 vs FO^2）、Prop 1.5（CoR vs FO^2）、Thm 1.6（ARA(3)）、Prop 1.7（RTC）は p.4 で結果の名前だけ
- 追加：導入の FO^k の例（3-cycle、推移性、closed walk）とモデル検査の表

補足：

- 投稿版の論文が確定したら定理番号を入れる（v8 の番号）：p.4 の定理 Thm 1.1、各項目 Thm 1.4・Prop 1.5・Thm 1.6・Prop 1.7、Key Lemma → Lemma 3.4・3.5、p.11 の "see the paper" → Lemma 3.5

k-variable logic FO^k

FO^k: first-order logic with k variables. For example, in FO^3:



$\varphi(x,y) := \exists z \left(E(x,y) \wedge E(y,z) \wedge E(z,x)\right)$ (a 3-cycle)

FO^3 also expresses, e.g.,

- transitivity: $\forall x \forall y \forall z \left(E(x,y) \wedge E(y,z) \to E(x,z)\right)$,
- the existence of a closed walk of length 4:
  $\exists x \exists y \left(E(x,y) \wedge \exists z \left(E(y,z) \wedge \exists y \left(E(z,y) \wedge E(y,x)\right)\right)\right)$ (any length ℓ, by reusing y and z).

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

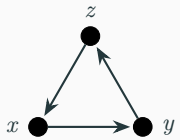
☞ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]
E.g., $O(n^3)$ for φ above: try all x, y, z

ページ概要：

- FO^k とモデル検査の問題設定
- 素直に計算すると $O(|\varphi| \cdot n^k)$。例の 3-cycle なら $O(n^3)$（次のページで n^ω に下がる）

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FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under Hyperclique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ \cdots \wedge \cdots$, we use DNF translation.
$\rightsquigarrow$ $2^{O(|\varphi|)}$ compositions are used [Nak22, IEAMP].

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- $k \geq 4$ では n^k が最適だが、FO^3 だけは CoR への翻訳で n^ω まで下がる
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Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

$$\rightsquigarrow_{\text{(to CoR)}}\ \Big(\bigcup_{A \subseteq [k]} \Big(\bigcap_{i \in A} P_i \Big) ; \Big(\bigcap_{i \notin A} Q_i \Big) \Big)(x,y)$$

(This translation uses 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 *to* CoR*?*

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

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Outline

1. Our results

2. Technical overview

3. Conclusion

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR*.*
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions, even with shared subterms.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary
  before: exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
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- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with RTC (reflexive-transitive closure)

👉 Our proof technique applies to all of them!

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- 答えは YES：$|\varphi_k| = O(k)$ の式に、合成 $\geq 2^{k-1}$ 個が必要
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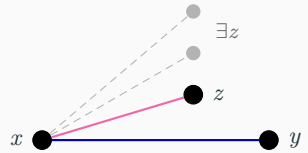
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The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



$G_k(x,y)$ holds iff there is z such that
for every $i \in [k]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order.

CoR: 2^k compositions (by DNF, as in the example before)

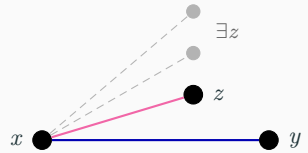
$$G_k \equiv \bigcup_{A \subseteq [k]} \left[\left(\bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left(\left(\bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$

ページ概要：

- 下界を与える式 G_k と、その DNF が合成 2^k 個であること（2 倍を除いて最適）

The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



$G_k(x,y)$ holds iff there is z such that
for every $i \in [k]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order.

CoR: 2^k **compositions** (by DNF, as in the example before)

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[\left(\bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left(\left(\bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$

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Outline

1. Our results

2. Technical overview

3. Conclusion

Proof outline (high level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

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G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
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☞ If $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

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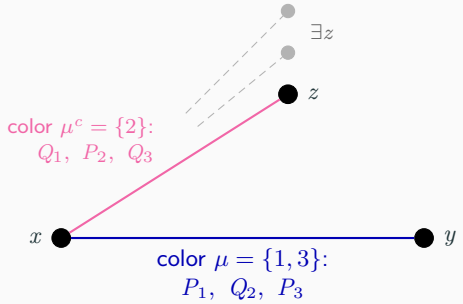
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Structures as colored complete graphs

View structures as **complete graphs**, colored with 2^k colors $\mu \subseteq [k]$: $P_i \iff i \in \mu$.

Q_i is used as the complement of P_i (i.e., $i \notin \mu$): exactly one of P_i, Q_i holds on each edge.



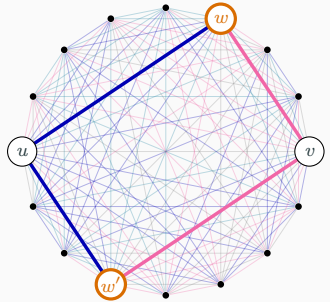
for every $i \in [3]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order

☛ $G_k(x, y)$ holds iff some x-z edge has the color μ^c, where μ is the color of the x-y edge.

ページ概要：

- 構造を色付き完全グラフとして見ると、G_k は色の言葉で読める

The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k4^k)$ vertices,
cf. [JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** w such that the u-w and w-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

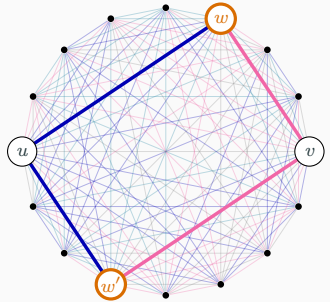
Every CoR term t is color-invariant on $\mathcal{M}_0$: $\mathcal{M}_0 \models t(u,v) \iff \mathcal{M}_0 \models t(u',v')$ whenever the u-v and u'-v' edges have the same color.

Proof by induction on t: the case $t = t_1 \,;\, t_2$ follows from the (two-)witness property.

ページ概要：

- ランダム彩色の $\mathcal{M}_0$ は two-witness property をもつ
- そのため、どの項の真偽も辺の色だけで決まる（color-invariant）

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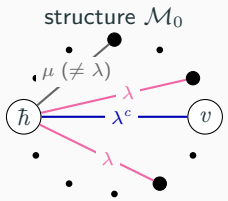
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$\mathcal{M}_\lambda$: flipping at the hub $\hbar$

Hub $\hbar$: a fixed arbitrary vertex of $\mathcal{M}_0$.



$\hbar$ has λ-edges

For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
Flipping each of the 2^k colors λ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.
Definition r **distinguishes** $\mathcal{M}_\lambda$ from $\mathcal{M}_0$ if $\mathcal{M}_0 \models r(u, v)$ but $\mathcal{M}_\lambda \not\models r(u', v')$ (or vice versa) for some u-v edge in $\mathcal{M}_0$ and u'-v' edge in $\mathcal{M}_\lambda$ of the same color.
$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slides).
(Boolean operations and converse act pair by pair, so they newly distinguish none.)

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Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (1/2)

Q. If t and s do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$, when can $t\,;s$?

Case $\hbar \notin \{u, v\}$: Never. ($t\,;s$ cannot distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$ at such (u, v).)

Similarly for the converse direction (from $\mathcal{M}_\lambda$ to $\mathcal{M}_0$).

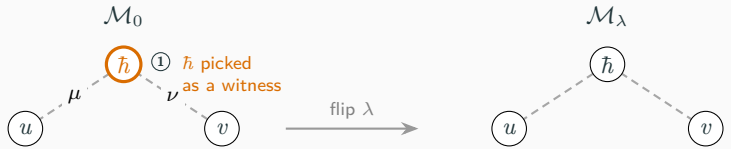
ページ概要：

- hub を含まない点対では、witness を同じ色の組の別の頂点に取り直せる
- よって合成はそこでは見分けない

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Similarly for the converse direction (from $\mathcal{M}_\lambda$ to $\mathcal{M}_0$).

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② The **two-witness property** shows there is w with the same colors (μ, ν)

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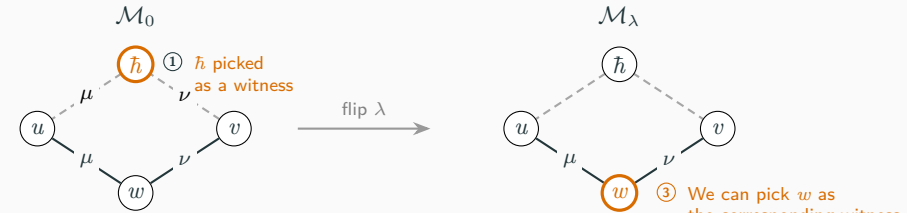
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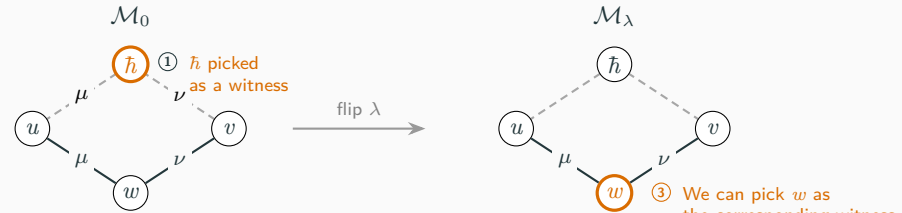
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Case $u = \hbar$ (similarly for $v = \hbar$): the $\hbar$-w edge itself may be flipped.

So the answer changes only if t holds on some λ-edges and only on them ($v = \hbar$: the same for s).
Such λ is uniquely determined for t and for s.
$\rightsquigarrow$ Each composition newly distinguishes at most two $\mathcal{M}_\lambda$.
(The remaining cases: the pair $(\hbar, \hbar)$ and witnesses $w \in \{u, v\}$. See the paper for details.)

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Outline

1. Our results

2. Technical overview

3. Conclusion

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Thm.

There are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Our proof also applies to FO^3 vs. FO^2, matrix query languages, and CoR with RTC.

Key Idea the randomly colored structure $\mathcal{M}_0$ with its 2^k flipped structures.
Each composition can newly distinguish ≤ 2 of them.

Problem (Open Questions)

- *a superpolynomial gap over a* ***fixed*** *finite signature (e.g., one binary relation)?*
- *the domain size from which the blowup begins?* *(known: between* $2^k/(8k)$ *and* $O(k4^k)$*)*
- *the exact number of compositions for* G_k*: between* 2^{k-1} *and* 2^k*?*

Thank you for your attention!

ページ概要：

- 結果（合成 $\geq 2^{k-1}$）と鍵となるアイデア（$\mathcal{M}_0$ の 2^k 通りの flip）のまとめ
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